# Minh-Quan Viet Bui

Ph.D. Candidate · Computer Vision · Neural Rendering · 3D/4D Reconstruction

*VIC Lab, KAIST, Daejeon, Republic of Korea*

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*"Building Efficient and Generalizable Methods for 3D/4D Scene Understanding"*

## Research Statement

I am a Ph.D. candidate at KAIST (VIC Lab, advised by Prof. Munchurl Kim) specializing in 3D computer vision, neural rendering, and 3D/4D scene reconstruction. My research builds **efficient, robust, and generalizable** representations of 3D/4D scenes from images and videos, spanning from per-scene optimization to feed-forward, generalizable approaches with applications in AR/VR, autonomous systems, computational photography, world modeling, and robotics.

My work has been published at **CVPR 2025 & 2026 (Highlight)**, **ECCV 2026**, **AAAI 2026**, and **IEEE TPAMI** (IF: 20.8). I am actively seeking **research internship, or postdoctoral positions** in 3D computer vision and world modeling.

## Education

**Korea Advanced Institute of Science and Technology (KAIST)** — *Daejeon, Republic of Korea*

Ph.D. Candidate, Electrical Engineering — *Aug. 2022 – Present*

## Publications

** Equal contribution / co-first authorship.* **9 first/co-first-author publications** (4 top-tier conferences, 5 journals).

### Conference Papers

[C1] **Minh-Quan Viet Bui***, Jongmin Park*, Juan Luis Gonzalez Bello, Jaeho Moon, Jihyong Oh, and Munchurl Kim. "EcoSplat: Efficiency-controllable Feed-forward 3D Gaussian Splatting from Multi-view Images." *CVPR, 2026.* **(Highlight)**

[C2] **Minh-Quan Viet Bui***, Jaeho Moon*, and Munchurl Kim. "AirSplat: Alignment and Rating for Robust Feed-Forward 3D Gaussian Splatting." *ECCV, 2026.*

[C3] **Minh-Quan Viet Bui***, Jongmin Park*, Juan Luis Gonzalez Bello, Jaeho Moon, Jihyong Oh, and Munchurl Kim. "MoBGS: Motion Deblurring Dynamic 3D Gaussian Splatting for Blurry Monocular Video." *AAAI, 2026.*

[C4] Jongmin Park*, Jeonghwan Yun*, **Minh-Quan Viet Bui**, Munchurl Kim. "WebSpline: Structure-Informed Splines for Real-Time 3D Gaussians from Monocular Videos." *NeurIPS, 2026.*

[C5] Jongmin Park*, **Minh-Quan Viet Bui***, Juan Luis Gonzalez Bello, Jaeho Moon, Jihyong Oh, and Munchurl Kim. "SplineGS: Robust Motion-Adaptive Spline for Real-Time Dynamic 3D Gaussians from Monocular Video." *CVPR, 2025.*

### Journal Papers

[J1] **Minh-Quan Viet Bui***, Jongmin Park*, Jihyong Oh, and Munchurl Kim. "MoBluRF: Motion Deblurring Neural Radiance Fields for Blurry Monocular Video." *IEEE Trans. Pattern Anal. Mach. Intell.* ***(TPAMI)***, pp. 1–18, 2025, doi: 10.1109/TPAMI.2025.3574644. **[IF: 20.8]**

[J2] Juan Luis Gonzalez Bello*, **Minh-Quan Viet Bui***, and Munchurl Kim. "ProNeRF: Learning Efficient Projection-Aware Ray Sampling for Fine-Grained Implicit Neural Radiance Fields." *IEEE Access*, vol. 12, pp. 56799–56814, 2024, doi: 10.1109/ACCESS.2024.3390753. **[IF: 3.4]**

[J3] Phan Thi Huyen Thanh*, **Minh-Quan Viet Bui***, Duc Dung Nguyen, Tran Vu Pham, Truong Vinh Truong Duy, and Natori Naotake. "Transfer Multi-Source Knowledge via Scale-Aware Online Domain Adaptation in Depth Estimation for Autonomous Driving." *Image Vis. Comput.*, vol. 141, p. 104871, 2024, doi: 10.1016/j.imavis.2023.104871. **[IF: 4.2]**

[J4] Duc Tuan Ngo*, **Minh-Quan Viet Bui***, Duc Dung Nguyen, and Hoang-Anh Pham. "eGAC3D: Enhancing Depth-Adaptive Convolution and Depth Estimation for Monocular 3D Object Pose Detection." *PeerJ Comput. Sci.*, vol. 8, p. e1144, 2022, doi: 10.7717/peerj-cs.1144. **[IF: 3.8]**

[J5] **Minh-Quan Viet Bui***, Duc Tuan Ngo*, Hoang-Anh Pham, and Duc Dung Nguyen. "GAC3D: Improving Monocular 3D Object Detection with Ground-Guide Model and Adaptive Convolution." *PeerJ Comput. Sci.*, vol. 7, p. e686, 2021, doi: 10.7717/peerj-cs.686. **[IF: 3.8]**

# Research Experience

**Ph.D. Researcher** — *Daejeon, Republic of Korea*

VIC Lab, KAIST (Advisor: Prof. Munchurl Kim) — *Mar. 2022 – Present*

- **AirSplat** (*ECCV 2026*): a robust feed-forward 3D Gaussian Splatting framework that adapts 3D vision foundation models for pose-free novel view synthesis; resolves pose-geometry misalignment with a self-consistent pose-alignment feedback loop and rating-based opacity matching that filters degraded primitives.
- **EcoSplat** (*CVPR 2026* ***Highlight***): a feed-forward 3D Gaussian Splatting framework where users specify any target Gaussian count at inference, with no per-scene retraining; best quality across efficiency budgets on RE10K and ACID.
- **MoBGS** (*AAAI 2026*): a motion-deblurring dynamic 3D Gaussian Splatting method for blurry monocular video using neural-ODE camera trajectory estimation and physical exposure modeling; $\sim$**12% better reconstruction** than prior SOTA on Stereo Blur at **480 FPS**.
- **SplineGS** (*CVPR 2025*): a spline-based dynamic 3D Gaussian Splatting framework for real-time rendering from monocular video, eliminating COLMAP preprocessing; **400 FPS** on the NVIDIA dataset.
- **MoBluRF** (*IEEE TPAMI 2025, IF: 20.8*): a two-stage motion-deblurring NeRF for blurry monocular video, jointly correcting camera poses and decomposing camera/object motion blur; state-of-the-art on the self-introduced Blurry iPhone benchmark and Stereo Blur.
- **ProNeRF** (*IEEE Access 2024, IF: 3.4*): a projection-aware ray sampling network for NeRF that focuses queries on high-density 3D regions; **15–23**$\times$ **faster** than vanilla NeRF with **+0.95 dB** over HyperReel.

**Research Assistant** — *Ho Chi Minh City, Vietnam*

Ho Chi Minh City University of Technology (HCMUT-VNU) — *2019 – 2021*

- **Cross-Domain Depth Estimation** (*Image & Vision Computing 2024, IF: 4.2*): a meta-learning online domain adaptation framework for monocular depth estimation; state-of-the-art on KITTI Eigen without target-domain supervision.
- **GAC3D / eGAC3D** (*PeerJ Computer Science 2021 & 2022, IF: 3.8*): monocular 3D object detection networks using ground-guide geometric priors and depth-adaptive convolutions for accurate 3D bounding-box estimation from single images.

# Invited Talks

**"EcoSplat: Efficiency-controllable Feed-forward 3D Gaussian Splatting from Multi-view Images"** — *IEEE/CVF CVPR 2026, Denver, CO*

NVIDIA Booth Spotlight Talk — *June 2026*

# Technical Skills

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| **Deep Learning** | PyTorch, JAX, CUDA, Python |
| **Neural Rendering** | 3D Gaussian Splatting (3DGS), Neural Radiance Fields (NeRF), Differentiable Rendering, Point Cloud Processing |
| **Computer Vision** | Dynamic Scene Reconstruction, Novel View Synthesis, Depth Estimation, 3D Object Detection |
| **Tools & Platforms** | Git, Docker, Linux (Ubuntu), MATLAB, LaTeX |

# Honors & Awards

2026 **Challenge Winner,** PhysAI Dynamic 4D Reconstruction Challenge, ECCV Workshop

2026 **Outstanding Reviewer,** CVPR

2022– **KAIST Graduate Program Scholarship,** Full tuition waiver and monthly living stipend — *KAIST*